5.3 Functions and Relations

Definition of a Relation: A set of ordered pairs (x, y) is called a relation in x and y.
The set of ***x* values** in the ordered pairs is called the **domain** of the relation.
The set of ***y* values** in the ordered pairs is called the **range** of the relation.

Definition of a Function: Given a relation in x and y, we say that ***y* is a function of *x*** if for each value of x in the domain, there is exactly one value of y in the range.

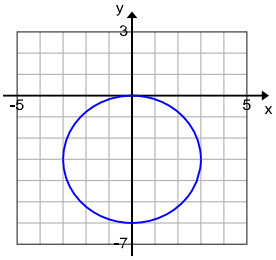
Example 4 Determine if the relation defines y as a function of x. Then write the domain and range.

(a) $\{(2,3),(2,4),(3,5)\}$ (b) $y^2 - x = 7$ (c) $y^3 - x = 7$

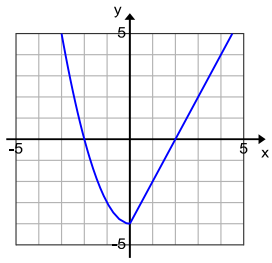
Vertical Line Test: A graph defines y as a function of x if no vertical line intersects the graph in more than one point.

Example 5 Determine if the graph defines y as a function of x.

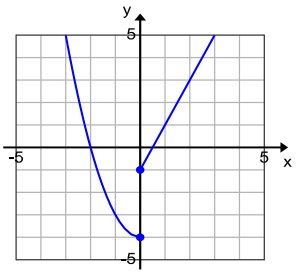
(a)



(b)



(c)



Function Notation

Consider the function $y = x^2 - 2$. Suppose we give this function the name f. Then, in place of y we use the notation $f(x)$, read "f of x," and we write $f(x) = x^2 - 2$. So, $y = f(x)$, meaning that $f(x)$ is just another name for y. x is the input variable, and $f(x)$ is the corresponding output, or y-value. Naming a function makes some things simpler. For example, if you are dealing with more than 1 function, you can refer to each function by name. Also, if you want to know the y value that corresponds to the x-value of 3, you can simply ask for $f(3)$.

Example 6 Let $k = \{(-1,-3),(6,5),(-3,0),(4,6)\}$. Find c.

(a) $c = k(-3)$ (b) $c = k(5)$ (c) $k(c) = 0$ (d) $k(c) = 6$

ExC: Let $f(x) = 4x^2 - 3x + 1$.
Write the function like this: $f(x) = 4(x)^2 - 3(x) + 1$. Whatever you input for x, you put in the parentheses.

$$f(-2) = 4(-2)^2 - 3(-2) + 1 \qquad = 23$$

$$f(h) = 4(h)^2 - 3(h) + 1 \qquad = 4h^2 - 3h + 1$$

$$f(x+3) = 4(x+3)^2 - 3(x+3) + 1 \qquad = 4x^2 + 21x + 28$$

Example 7 Let $f(x) = x^2 - x - 4$, $g(t) = 3 - |t|$ and $h(w) = \frac{2}{w}$ Find the following.

(a) $f(3)$

(b) $g(-3)$

(c) $h(\frac{4}{w+3})$

(d) $f(x+3)$

(e) $f(x+h)$

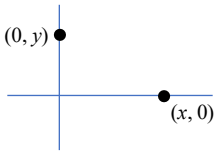
Your Turn 1: Find (a) $f(2)$, (b) $f(x-1)$, (c) $h(-\frac{3}{x+1})$

Example 8 (a) Suppose that a phone card has 400 minutes. Write a relationship that represents the number of minutes remaining $r(x)$ as a function of the number of minutes already used *x*.

(b) Write a relationship for a function whose *f(x)* values are 2 more than 5 times the square root of *x*.

Intercepts

For $y = f(x)$: The x-intercepts are the real solutions to the equation $f(x) = 0$.
The y-intercept is given by $f(0)$.

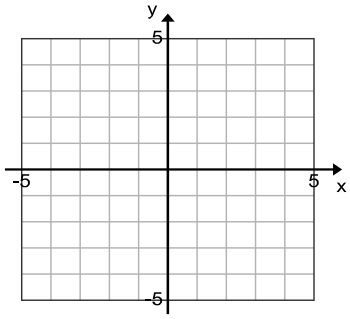


Example 9 Find the x-and y-intercepts and write them as ordered pairs.
Note: Denote what it is that you are deriving. For example, if finding x-intercepts, write *x-int:*

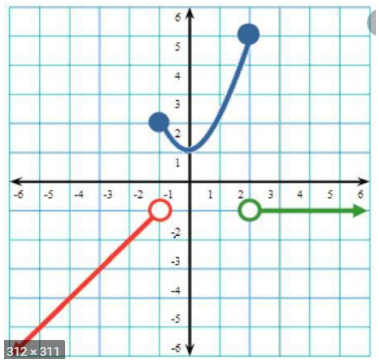
(a) $f(x) = x^2 - 4$ (b) $g(x) = 3 - |x|$ (c) $h(x) = \sqrt{x+1} - 2$

Graphing

Example 10 Graph $y - |x| = -1$ by plotting points. Determine the x and y intercepts.



Example 11 Use the graph of $y = f(x)$:



(a) $f(0) =$ (b) $f(2) =$ (c) $f(5) =$

(d) Find all x for which: (1) $f(x) = 2$ (2) $f(x) = -1$

(e) Solve (1) $f(x) > 0$ (2) $f(x) < 0$

(f) Identify the x-intercept: _____ y-intercept:_____

(g) Determine the domain of f _____ range of f _____

Domain

1. We can*not* take an even root of a negative number, therefore we must take the *inside* of the root and set it greater than or equal to zero.

2. We can*not* divide by zero, therefore we take the *entire* denominator and set it *not* equal to zero.

1. $\sqrt[n]{\geq 0}$ (*n* is even) ***INSIDE* of root ≥ 0**

2. $\frac{k}{\neq 0}$ ***DENOMINATOR* ≠ 0**

Example 12 Determine the domain for each function. Write your answers using interval notation.

(a) $f(x) = x + 3$

(b) $f(x) = \sqrt{x+3}$

(c) $f(x) = \frac{\sqrt{x+3}}{2x}$

(d) $g(x) = \frac{x-6}{\sqrt{x+3}}$

(e) $g(x) = \frac{x-6}{\sqrt{x+3}-1}$

(f) $g(x) = \frac{6}{|x+3|-1}$

(g) $g(x) = \frac{x-2}{x^2-3x-28}$

(h) $k(x) = \frac{\sqrt[3]{x-1}}{x^2+1}$

Your Turn 2: Find the domain. (a) $f(x) = \frac{\sqrt{x+5}}{2x-1}$ (b) $g(x) = \frac{\sqrt[3]{x}}{x^2-3x-4}$

5.4-5.5 Lines and Linear Applications

Forms of linear equations

Slope-intercept form: $y = mx + b$

Standard form: $Ax + By = C$

Slope formula

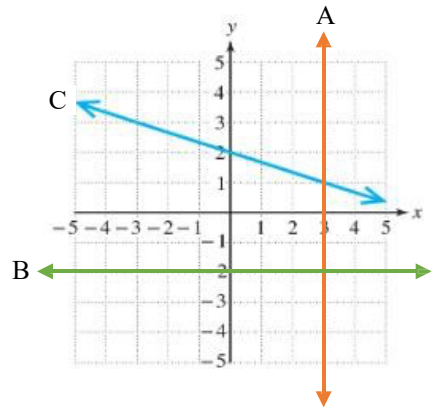
$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Point-slope formula for finding the equation of a line

$$y - y_1 = m(x - x_1)$$

Parallel lines have the same slope and ***perpendicular lines*** have slopes that are negative reciprocals of each other.

Example 13 Consider the graph of lines *A, B* and *C.*



(a) Find the slope of each line.

Line *A* _____

Line *B* _____

Line *C* _____

(b) Find the equation of each line.

Line *A* _____

Line *B* _____

Line *C* _____

Example 14 Find the equation of the line that:

Hint: Draw a picture!

(*i*) is parallel to the *x*-axis with *y*-intercept –6

(*ii*) has undefined slope and passes thru (1, -8)

(*iii*) has slope -2 and passes thru (0, 5)

Example 15 A line passes thru the points (2, -1) and (-4, 3). Find the equation of the line using

For either, we find slope first:

(a) $y = mx + b$

(b) $y - y_1 = m(x - x_1)$

Your Turn 3: Find the equation of the line passing through (1, -3) and (-1, 5)

Example 16 Find the equation of the line that passes thru (2, -3) and is perpendicular to $-4x + 2y = 3$.

Step 1: Write -4*x* + 2*y* = 3 in slope-intercept form and determine its slope. Use this to find the slope of the line in question.

Step 2: Use either $y = mx + b$ or $y - y_1 = m(x - x_1)$ to find the equation of the line in question.

Linear Cost, Revenue and Profit Functions

- **Cost** $C(x) = mx + b$ where m is the variable cost per item and b is the fixed cost.
- **Revenue** $R(x) = px$ where p is the price per unit
- **Profit** $P(x) = R(x) - C(x)$

Break-even point is when profit is 0 (cost equals revenue)

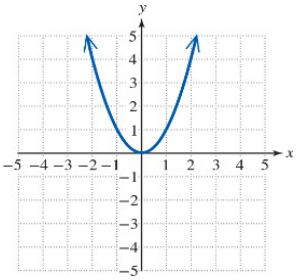
Example 18 At a parking garage, the charge for parking consists of a flat fee of \$2.00 plus \$1.50/hr. Write a linear function to model the cost of parking *P(t)* for *t* hours.

Example 19 At a summer art show a vendor sells lemonade for \$2.00 per cup. The cost to rent the booth is \$120. Furthermore, the vendor knows that the lemons, sugar and cups collectively cost \$0.50 for each cup of lemonade.
(a) Write a linear cost, revenue and profit function that models this situation.

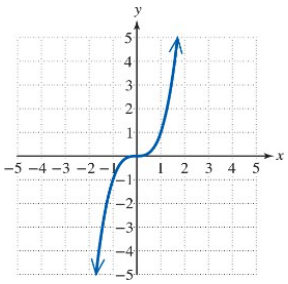
(b) Determine the break-even point.

6.1 Basic Graphs and Transformations

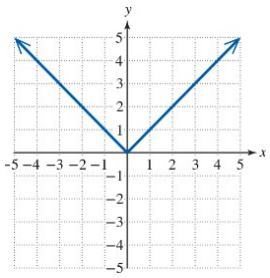
$f(x) = x^2$



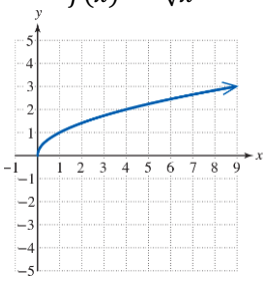
$f(x) = x^3$



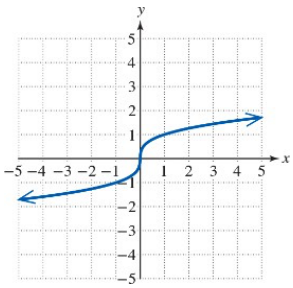
$f(x) = |x|$



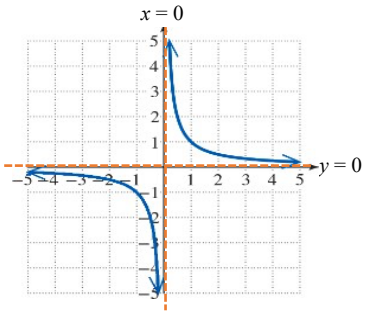
$f(x) = \sqrt{x}$



$f(x) = \sqrt[3]{x}$



$f(x) = \frac{1}{x}$



Transformations of *f(x)*

		Example: $y = \sqrt{x}$	
$af(x)$	stretch (or shrink) by a units	$y = 2\sqrt{x}$	stretch by 2
$-f(x)$	reflect about *x*-axis, denoted R_x	$y = -\sqrt{x}$	R_x
$f(-x)$	reflect about *y*-axis, denoted R_y	$y = \sqrt{-x}$	R_y
$f(x) + k$	shift up/down by $\lvert k \rvert$ units	$y = \sqrt{x} + 3$	shift 3 up
$f(x - h)$	shift left/right by $\lvert h \rvert$ units	$y = \sqrt{x + 3}$	shift 3 left

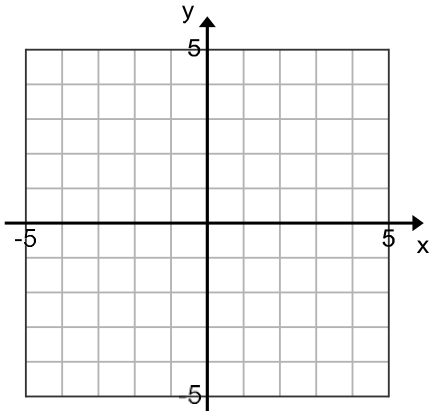
Transformations of Functions

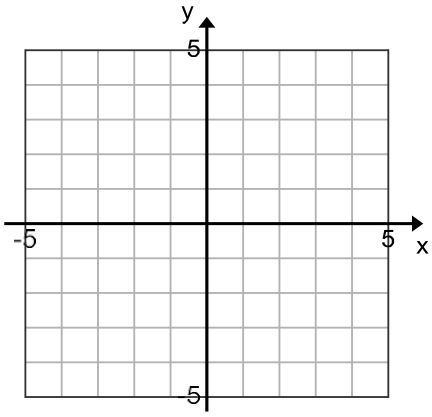
Example 20 Determine the transformation being applied to the basic curve.
(a) $y = 3|x|$ (b) $y = -\frac{1}{x}$ (c) $y = \frac{1}{x} - 3$ (d) $y = \frac{1}{x-3}$ (e) $y = \sqrt[3]{-x}$

Example 21 Graph $f(x) = 3 - 2(x + 1)^2$ by applying transformations in the indicated stages. Indicate 3 points on each curve.

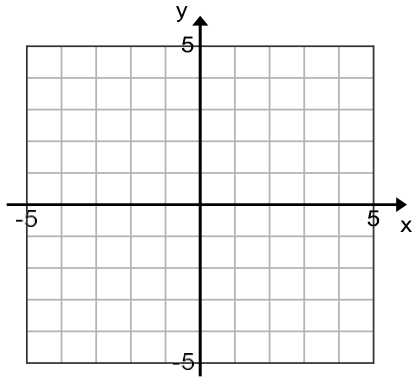
Basic curve Stretch/Shrink Reflection Shifts: _____

$y =$ _____ $y =$ _____ $y =$ _____ $y = 3 - 2(x + 1)^2$

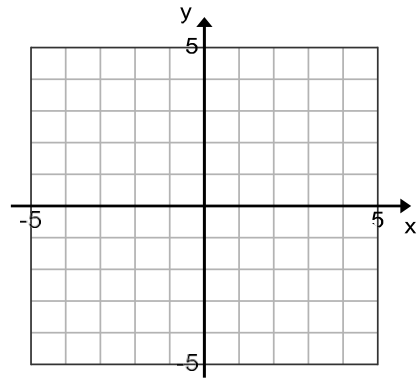




Example 22 Graph (a) $y = \sqrt{3 - x} - 1$



(b) $y = \frac{1}{x-1} + 2$

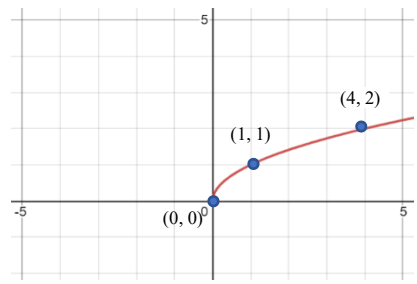


Your Turn 4: Do as we did in **Ex21** with $f(x) = -\sqrt{x+1} + 3$

Transformations of Basic Curves: ***Graphing Procedure***

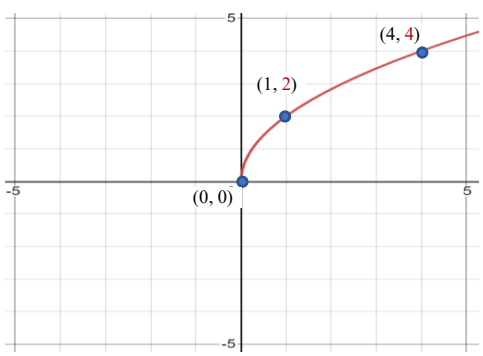
ExD: Graph $f(x) = 1 - 2\sqrt{x+3}$

Step 1: Graph the "parent" function: $y = \sqrt{x}$



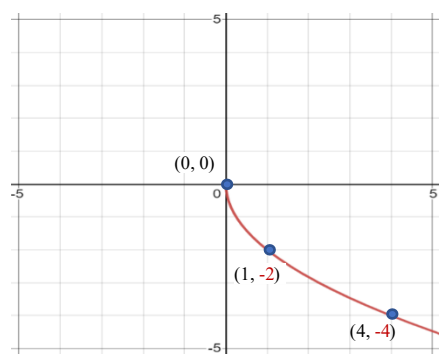
Step 2: Pick out any reflections and stretching:

Stretch by a factor of 2: $y = 2\sqrt{x}$



Each y-value is twice what it was in the previous graph.

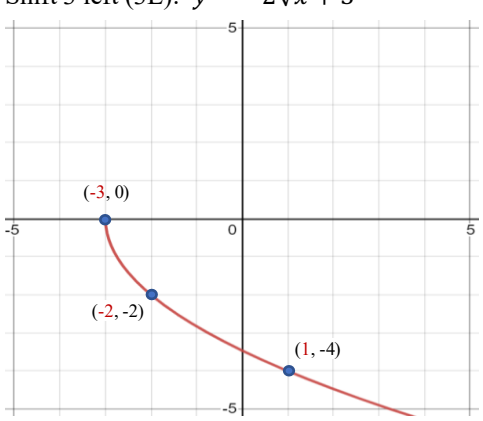
Reflection about the x-axis (R_x): $y = -2\sqrt{x}$



Each y-value is the negative what it was in the previous graph.

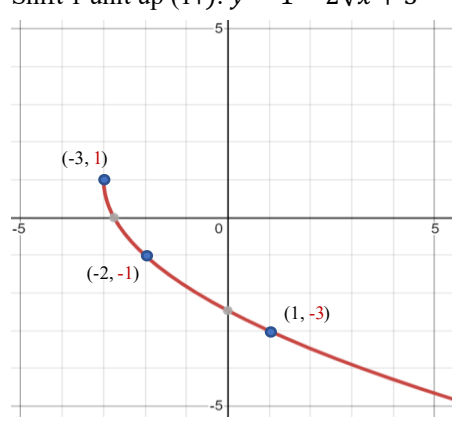
Step 3: Pick out any shifts:

Shift 3 left (3L): $y = -2\sqrt{x+3}$



Each x-value is 3 less than what it was in the previous graph.

Shift 1 unit up (1↑): $y = 1 - 2\sqrt{x+3}$



Each y-value is 1 more than what it was in the previous graph.

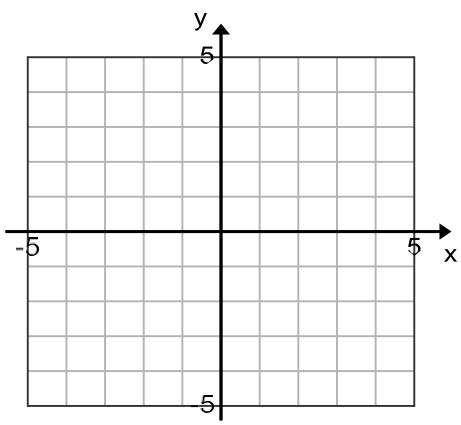
6.2 Piecewise Defined Functions

Example 23 Let $f(x) = \begin{cases} -2x-3 & for \quad x < -2 \\ x^2+1 & for \ -2 \le x < 1. \\ 2 & for \quad x > 1 \end{cases}$ (a) evaluate: $f(0) =$ $f(-5) =$

$f(1) =$ $f(4) =$

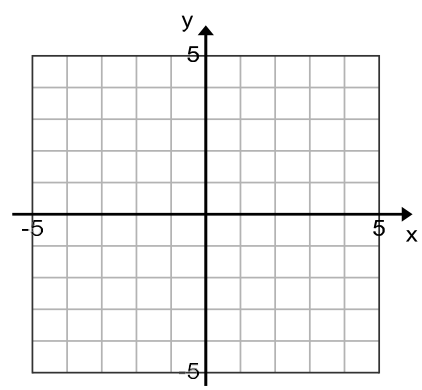
(b) Graph *f*.

1. $y = -2x - 3, x < -2$ 2. $y = x^2 + 1, \ -2 \le x < 1$ 3. $y = 2, \ x > 1$

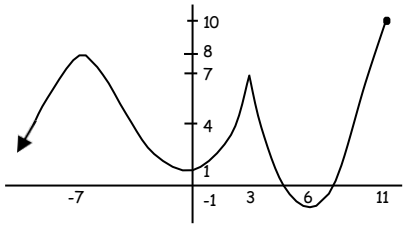


.**Example 24B**

Your Turn 5: Graph $f(x) = \begin{cases} (x+2)^2 & if \quad x < -2 \\ -3 & if \ x > -2 \end{cases}$



6.3 Increasing, Decreasing and Constant Intervals; Min and Max Values of a Function



f is *increasing* over the *x*-intervals (-∞, -7), (0, 3) and (6, 11)
f is *decreasing* over the *x*-intervals (-7, 0) and (3, 6)

Relative *maxima* of **8** and **7** occur at $x = -7$ and 3, respectively.
Relative *minima* of **1** and **-1** occur at $x = 0$ and 6, respectively.

Endpoint values do not count as relative minima or maxima.

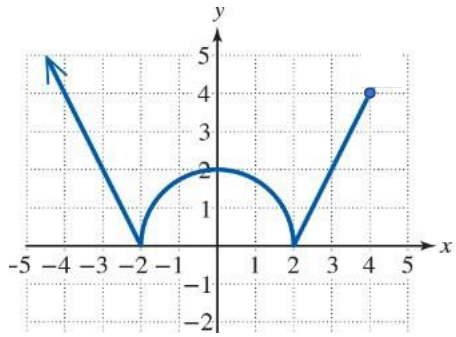
Example 26 Use interval notation to write the interval(s) over which $f(x)$ is increasing, decreasing or constant. Identify the value and location of any relative maxima or minima of the function.

Increasing: _____

Decreasing: _____

Constant: _____

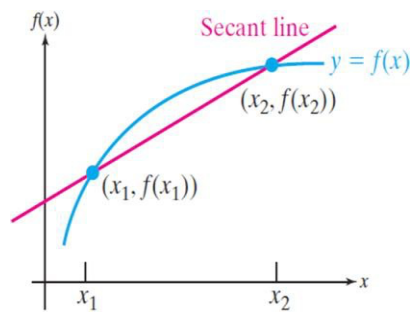
Relative minima: Relative maxima:



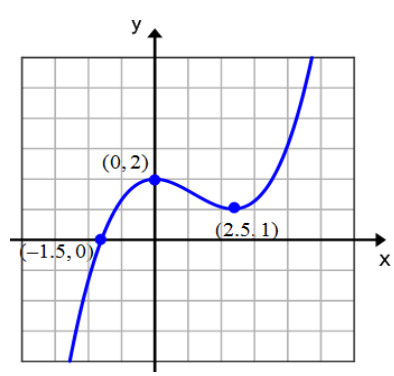
6.3 Average Rate of Change (*ARC*) and the Difference Quotient

If *f* is defined on the interval $[x_1, x_2]$, then the **average rate of change** of *f* on the interval $[x_1, x_2]$ is the *slope of the secant line* containing points $(x_1, f(x_1))$ and $(x_2, f(x_2))$.

Average rate of change: $\frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$ or $m_{secant} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$

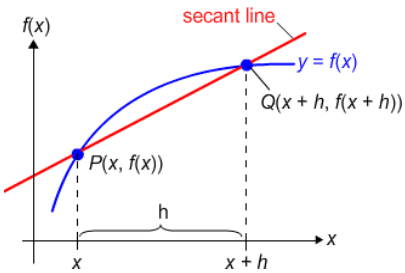


Example27 Find the average rate of change of the graphed function *f* over the interval [-1.5, 2.5]. Use function notation. Draw the graphical representation of this *ARC* on the graph.



Example28 Determine the *ARC* of $f(x) = 2x^2 + 1$ on the interval [-2, 3]. Show the graphical representation.

The Difference Quotient



The average rate of change between P and Q is the slope of the secant line and is given by:

$$m = \frac{\Delta y}{\Delta x} = \frac{f(x+h) - f(x)}{(x+h) - x} = \frac{f(x+h) - f(x)}{h}$$

The expression $\dfrac{f(x+h) - f(x)}{h}$ is called the difference quotient.

Example30 Let $f(x) = 2x^2 - x + 1$ and $g(x) = \frac{5}{x+6}$. Evaluate each difference quotient.

a) $\frac{f(x+h)-f(x)}{h}$

b) $\frac{g(x+1)-g(1)}{h}$

Your Turn 6: $f(x) = 2x^2 - x + 1$. Evaluate $\frac{f(2+h)-f(2)}{h}$

6.4 Combining and Composing Functions

Combining Functions

$(f+g)(x) = f(x) + g(x)$	The domain of each is all x that works for *both* functions f and g.
$(f-g)(x) = f(x) - g(x)$	
$(fg)(x) = f(x)g(x)$	
$(\frac{f}{g})(x) = \frac{f(x)}{g(x)}$	The function f/g has the added restriction that $g(x) \neq 0$.

Example31 Let $f(x) = -2x^2 + 7$ and $g(x) = 2x - 8$. Find the following.

(a) $(f-g)(x)$

(b) $(\frac{f}{g})(2)$

Example32 Let $f(x) = \frac{1}{x-1}$ and $g(x) = \frac{x-2}{x^2-1}$. Find each function and its domain.

(a) $(f+g)(x)$

(b) $(\frac{g}{f})(x)$

<u>Composing Functions</u>: $(fog)(x) = f[g(x)]$

The domain of fog is all the x values in the domain of the "inner" function, g that *also work* for the end result.

When we compose two previously defined functions, let's say f and g, we create a new function that we name fog, read "f of g." The name fog is simply notation for $f[g(x)]$, meaning that we are merely evaluating the function f at the function g. $g(x)$ is called the "inner function" because it is *inside* the function f. Be careful about the domain for composite functions. There are *two* places to check: (1) the domain of the "inner function" and (2) the end result. In summary,

- $(fog)(x) = f[g(x)]$
- $g(x)$ is called the "inner function" because it is *inside* the function f.
- The domain of fog is all the x values in the domain of the "inner" function, g that *also work* for the end result

ExE: Let $f(x) = \frac{3}{x^2-4}$ and $g(x) = \sqrt{x}$.

Then $(f \circ g)(x) = f[g(x)]$

$= f(\sqrt{x})$

$= \frac{3}{(\sqrt{x})^2 - 4}$

$= \frac{3}{x-4}$

So, $(f \circ g)(x) = \frac{3}{x-4}$.

$Domain_{f \circ g} : x \geq 0, x \neq 4$

The domain for this function must satisfy the domain of g $(x \geq 0)$, as well as the end result, which necessitates that $x \neq 4$.

NAME your function! When you combine or compose two functions, you are creating a new *function*, not just some answer, so you must include the *name* of the function on the left. In the above example, your "answer" is *not* $\frac{3}{x-4}$. You must write $(f \circ g)(x) = \frac{3}{x-4}$.

Example33 Let $f(x) = \frac{5}{x^2-1}$ and $g(x) = \sqrt{x+1}$. Find the following:

(a) Domain of f: _____

(b) Domain of g : _____

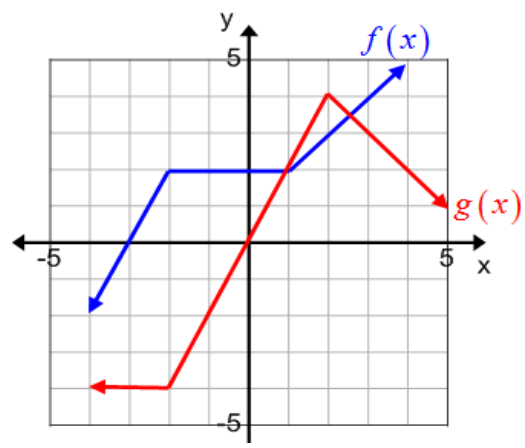
(a) $(fog)(x)$ and its domain

(d) $(gog)(3)$

Example34 Let $s(x) = \frac{5}{x+1}$ and $t(x) = \frac{2}{3x}$ Find $(s \circ t)(x)$ and its domain.

Your Turn 7: Let $f(x) = \frac{5}{x^2-1}$, $g(x) = \sqrt{x-2}$ and $h(x) = \frac{1}{\sqrt{x}}$ Find (a) $(fog)(x)$ and (b) $(foh)(x)$ their domain

Example35 Estimate the function values from the graph.



(a) $(f - g)(3)$

(b) $(\frac{g}{f})(-2)$

(c) $(g \circ f)(1)$

(d) $(f \circ f)(-3)$

Your Turn 8: Use the graph above to evaluate (a) $(f \circ g)(1)$ (b) $(g + f)(2)$

9.1 Inverse Functions

An inverse function is *not* a reciprocal function. It is a function that does the ***inverse*** of another function. Some nonmathematical examples are:

1. Function: Tying your shoe. Inverse function: Untying your shoe
2. Function: Sitting down Inverse function: Standing up

A mathematical example of a pair of inverse functions would be $y = x^3$ and $y = \sqrt[3]{x}$. The one function cubes things, while the other function *un*cubes them. Not all functions have inverses. For example, an equation for the inverse relation of $y = 3x^2 + 2$ is given by $x = 3y^2 + 2$. However, if you try to solve the second equation for y you will obtain $y = \pm\sqrt{\frac{x-2}{3}}$, which is *not* a function.

Function: Each x corresponds to exactly one y. A function passes a "**Vertical Line Test**."
One-to-One (1-1) Function: A function where each y corresponds to exactly one x. A 1-1 function passes a "**Horizontal Line Test**."
Horizontal Line Test: If each *horizontal* line intersects the curve exactly once, then we have a 1-1 function.
Fact: A function f has an inverse if and only if f is a 1-1 function.

Finding an Inverse

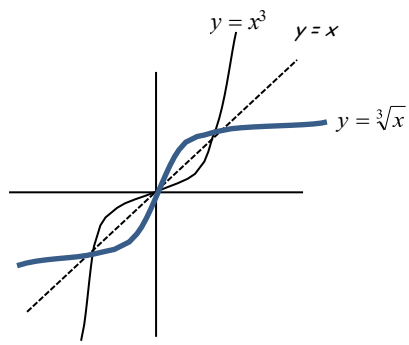
The function $y = x^3$ is one-to-one.

To find its inverse:

1. Exchange x and y: $x = y^3$
2. Solve for y: $\sqrt[3]{x} = y$

$y = \sqrt[3]{x}$

Graphing Inverses



Composing Inverses

Let $f(x) = x^3$ and $g(x) = \sqrt[3]{x}$.

1. $(fog)(x) = f(g(x))$
$= f(\sqrt[3]{x})$
$= (\sqrt[3]{x})^3$
$= x$

2. $(gof)(x) = g(f(x))$
$= g(x^3)$
$= \sqrt[3]{x^3}$
$= x$

Inverse Functions: f and g are inverse functions if and only if
$(fog)(x) = x$ and $(gof)(x) = x$

Therefore, $f(x) = x^3$ and $g(x) = \sqrt[3]{x}$ are inverse functions. Using *inverse notation*, if $f(x) = x^3$, then $f^{-1}(x) = \sqrt[3]{x}$.

Inverse Notation: f^{-1} denotes the inverse of f. $f^{-1} \neq \frac{1}{f}$

Properties: Because of the swapping of *x* and *y*:

(a) If $f(a) = b$, then $f^{-1}(b) = a$. Which also means if (a, b) is on the graph of f. then (b, a) is on the graph of f^{-1}.

(b) D_f becomes $R_{f^{-1}}$ and R_f becomes $D_{f^{-1}}$ **Ex:** If the domain of f is $-3 < x < 10$, then the range of f^{-1} is $-3 < y < 10$.

(c) x-int of f becomes y-int of f^{-1} and y-int of f becomes x-int of f^{-1}. **Ex:** If f has x-int (2, 0), then f^{-1} has y-int (0, 2).

(d) VA of f becomes HA for f^{-1} and HA of f becomes VA for f^{-1}. **Ex:** If f has a VA of $x = 1$, then f^{-1} has a HA of $y = 1$.

Example 10 Function f is shown in the graph.

(a) Graph f^{-1} on the same set of axes.

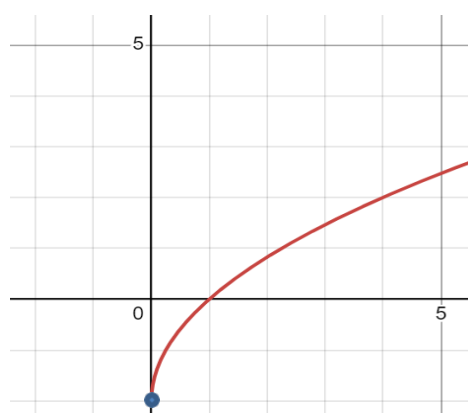
(b) $f(1) =$ ____, and f^{-1}(___) = ___

(c) _____lies on the graph of f, and

_____lies on the graph of f^{-1}.

(d) The domain of f is _____, and

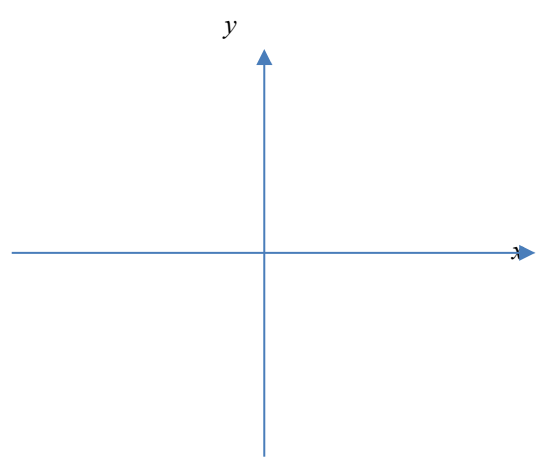
the range of f^{-1} is _____.



Example 11 Let $f(x) = \frac{2x+4}{x-1}$. Find $f^{-1}(x)$ and range of inverse function.

Example 12 Let $f(x) = 2\sqrt{x+9}$ Find $f^{-1}(x)$ and domain of inverse function.

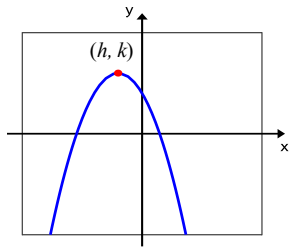
Example 13 Let $f(x) = (x+3)^2 - 1, x \geq -3$. Find $f^{-1}(x)$ and then graph both on the same set of axes.



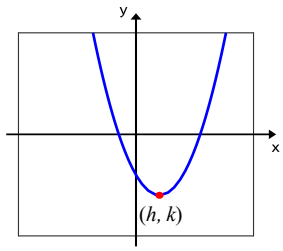
7.1 Quadratic Functions

The graph of the quadratic $f(x) = ax^2 + bx + c$ is a parabola and may be written in vertex form as $f(x) = a(x-h)^2 + k$.

$a < 0$



$a > 0$



Vertex Form $f(x) = a(x-h)^2 + k$

1. If a > 0, parabola opens UP. If a < 0, parabola opens DOWN.
2. Vertex: (h, k)
3. Axis of Symmetry: x = h
4. k is the minimum value if $a > 0$ and the maximum value if $a < 0$
5. Range: $(-\infty, k]$ if a < 0 and $[k, \infty)$ if a > 0

Converting to Vertex form $f(x) = a(x-h)^2 + k$

Example 36 (a) Write $f(x) = -2x^2 + 8x - 2$ in vertex form.

Step 1: Isolate x-terms by putting a parenthesis around them: $f(x) =$

Step 2: Factor out a from the x terms: $f(x) =$

Step 3: We are completing the square on the x-terms, $x^2 + bx$. We add $(\frac{1}{2}b)^2$ to the x-terms, and then add/subtract the necessary value in order to compensate for the addition of $(\frac{1}{2}b)^2$. The right side of the function must have a *NET GAIN of ZERO!*

$(\frac{1}{2}b)^2 =$ $f(x) =$

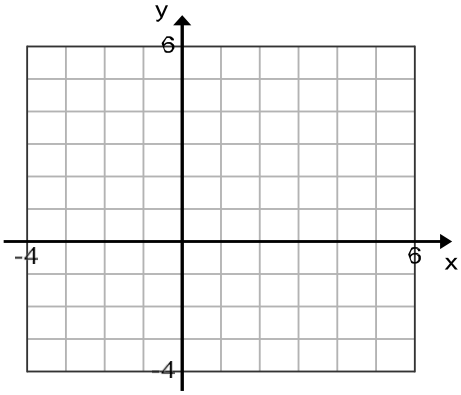
(b) The vertex is _____. f opens UP / DOWN so f has a MIN / MAX value of _____located at _____.

The axis of symmetry is _____

(c) Find the x-intercepts.

(d) Find the y-intercept.

(e) Draw in the axis of symmetry. Then graph by plotting the vertex and other points that show the symmetry about its axis.



	x	y
vertex		
x-values 1 away from vertex		
x-values 2 away from vertex		
x-values 3 away from vertex		

(f) Domain: _____ Range: _____

Your Turn 9: Write $f(x) = 2x^2 + 4x - 6$ in vertex form by completing the square. Find the intercepts and graph.

Vertex: For $f(x) = ax^2 + bx + c$, the

x-coordinate of the vertex is $-\frac{b}{2a}$ and the

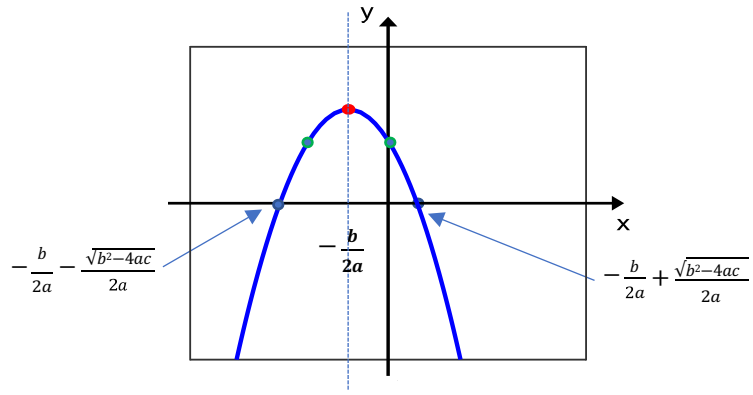
y-coordinate of the vertex is $f(-\frac{b}{2a})$.

Why?

The x-intercepts are solutions to $ax^2 + bx + c = 0$.

So, the x-intercepts are $x = -\frac{b}{2a} \pm \frac{\sqrt{b^2-4ac}}{2a}$.

Halfway between these two intercepts, at $x = -\frac{b}{2a}$, is the x-value of the vertex.



Example 37 Let $f(x) = 2x^2 - 8x + 6$.

(a) Find the x-value of the vertex using the given method.

1) x-intercepts:

2) $x = \frac{-b}{2a}$:

(b) Find the y-value of the vertex.

Vertex: _____

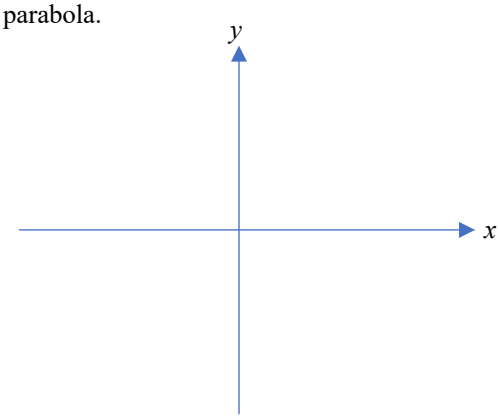
(c) Using the vertex found above, write the function in vertex form:

(d) f has a _____value of _____located at _____.

The axis of symmetry is _____.

Note: In the 2 forms $f(x) = ax^2 + bx + c$ and $f(x) = a(x-h)^2 + k$ the a-values are the same.

(e) Draw a sketch of the function. Include the axis of symmetry, the x-intercepts and two other points that show the symmetry of the parabola.



(f) Use the graph to solve the *non*linear inequality $2x^2 - 8x + 6 > 0$.

Your Turn 10: Let $f(x) = 2x^2 - 12x - 14$. Find the vertex by using (a) the x-intercepts, (b) $x = \frac{-b}{2a}$. Then write f in vertex form without completing the square.

Example 38 Find the quadratic function whose vertex is (-2, 3) and passes through the point (1, -3).

Quadratic Models; Maximizing or Minimizing Functions

Example 39 The monthly profit for a company that makes decorative picture frames depends on the price per frame. The company determines that the profit is approximated by $f(p) = -80p^2 + 3440p - 36{,}000$ where p is the price per frame and $f(p)$ is the monthly profit based on that price.

(a) Find the price that generates the maximum profit.
(b) Find the maximum profit.
(c) Find the price(s) that would enable the company to break even.

Example 41 Suppose that 60 meters of fencing is available to enclose a rectangular garden. The garden will be divided into three adjacent areas.

(a) Determine a function that models the area of the garden.

(b) Graph the function.

(c) Find the dimensions for the garden that will guarantee a maximum area, and also determine the maximum area.